iMED

An iPSC-based platform for developing personalized medicine for rare diseases (MED)

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1. General information

Title: **An iPSC-based platform for developing personalized medicine for rare diseases (iMED)**

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

*Name**	*Insti-tute*	*Department*	*Email*	*Position*	*EMC: clinician/ research/ mix*	*Role in Flagship*
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Planned start date of seed funding: *3-10-2022*
Primary Convergence theme: *Fundamentals of Health & Disease*
Secondary Convergence theme (*none*): Choose an item.
Keywords: *Rare disease, Personalized Medicine, Genetic therapies, iPSC, Organ-on-a-Chip, Nanotechnology, High Resolution Imaging*

1.b Summary (289/300 words)

Next-generation sequencing has revolutionized the diagnostic process of individuals suspected to suffer from a genetic disorder. **The identification of >1200 causal genes for both neurodevelopmental disorders and neuromuscular disorders provides unique pathophysiological insight and the opportunity to discover efficacious therapeutic targets.** In parallel, **innovative genetic therapies** are now moving towards clinical practice, in particular antisense oligonucleotide (AON) treatments and gene therapy using viral vectors. Importantly, such personalized treatments can readily be developed in an academic setting and offered at cost price.

These **exciting developments are accompanied by major challenges and force us to change our current methodology** in establishing the pathogenicity of the identified genetic variants, studying the underlying pathophysiology of these disorders, and identifying potential treatments. **Induced pluripotent stem cell (iPSC) technology** can fill this gap and **provide a powerful tool for developing personalized treatments.**

In the iMED Flagship, we have gathered a team of internationally leading TUD, EUR, and EMC scientists to address these challenges by **establishing a platform to study rare diseases, identify therapeutics and establish a path to bring them to the patient.** As a proof-of-principle, we will focus on **neurodevelopmental and neuromuscular disorders** for which the EMC partners have made major advances in developing genetic treatments. The overlapping technical demands for organ-on-a-chip and therapy development are addressed in collaboration with the TUD partners. Together with EUR partners we will address critical ethical and socioeconomic issues concerning first-in-human personalized treatments for these rare diseases.

Early expansion of iMED is expected to include the development of **an iPSC-based platform for assessing pathogenicity of genetic variants of unknown significance**, and development of **therapies for retinal disorders**. Taken together, this project has the potential to revolutionize the development of personalized medicine for the treatment of rare genetic diseases.

1.c Public Summary (281/300 words)

Met de opkomst van nieuwe diagnostische DNA-analyse methodes zijn de mogelijkheden om ziektegenen te identificeren in patiënten met (zeer) zeldzame aandoeningen enorm toegenomen. De identificatie van dergelijke DNA-veranderingen geeft inzicht in de fundamentele biologie van de ziekte en stelt ons in staat om onderliggende mechanismen te onderzoeken en therapieën te ontwikkelen.

Voor deze patiënten zijn therapieën gericht op correctie van de ziekteverwekkende DNA-veranderingen zeer veelbelovend. Met name genetische behandelingen die het effect van de mutatie ongedaan maken op DNA- of RNA-niveau bieden mogelijkheden voor gepersonaliseerde behandeling. Zo'n behandeling kan relatief eenvoudig in een academische setting ontwikkeld worden en tegen kostprijs aan de patiënt worden aangeboden. Met behulp van stamcellen (iPS-cellen) die uit patiëntmateriaal verkregen worden, kan weefsel van de patiënt gekweekt worden dat niet eenvoudig te verkrijgen is, zoals bijvoorbeeld hersen- en spierweefsel. Op deze gekweekte weefsels kan onderzocht worden wat voor effect de DNA-verandering heeft op het functioneren van de cellen en of genetische therapie mogelijk is. Deze technieken zijn echter zeer bewerkelijk, tijdrovend, wisselvallig en duur.

Grootschalige implementatie van deze technologische ontwikkelingen vereist een technologisch geavanceerd platform om zulke therapieën op snelle en relatief goedkope wijze te identificeren. In iMED willen we daarom als team van TUD- en EMC-experts een geautomatiseerd platform ontwikkelen om iPS-cellen van patiënten met zeldzame ziektes te verkrijgen en daarvan afgeleide weefsels te kweken. Daarnaast zullen we functionele testen ontwikkelen om gerichte, gepersonaliseerde therapieën te onderzoeken. In samenwerking met de EUR worden de ethische kwesties met betrekking tot gepersonaliseerde behandelingen onderzocht, evenals de economische evaluatie van zulke behandelingen. Het iMED Flaghip heeft de potentie om een leidende rol te spelen in de ontwikkeling en toepassing van gepersonaliseerde behandeling voor een breed scala (zeldzame) genetische aandoeningen.

2. Relevance to Convergence Health & Technology mission (50%) (2340/2500 words for 2.a to 2.e)

2.a Contribution to H&T mission and fit with H&T theme(s)

The iMED flagship will showcase the disciplinary convergence of EMC, TUD and EUR expertise, intertwining medical and life sciences, engineering, and social sciences capabilities to deliver a first-of-its kind platform for developing personalized medicines using patient-derived iPSCs.

Contribution to H&T mission

iMED will address a tremendous unmet need for patients with rare diseases, by developing genetic therapies such as targeting specific mutations using AONs or vector-based gene therapy. Since a given mutation could well be unique (private) to a single patient, sequence-specific **AON therapy is the ultimate example of individually tailored precision medicine**. Other genetic therapy strategies might be more broadly applicable to groups of patients with specific rare diseases (*e.g.* genetic therapies for SMA and Angelman Syndrome). The ability to restore protein function using gene therapy potentially allows for transformative therapy with **life-long health benefits from the moment of treatment to the end of life**. Once established, our platform will enable the identification of appropriate treatment within a year following diagnosis[1], which closely aligns with the mission of **prevention and early diagnostics to proactively maintain health**. To achieve **socio-economic equality in health(care)**, we will ensure that innovative therapies are available to all who might benefit from it. Therefore, the iMED flagship will create a platform to accelerate translation of fundamental knowledge of gene mutations into ethically acceptable, societally supported solutions to treat rare genetic diseases. iMED fully aligns with the ErasmusMC-Sophia focus areas of Rare Diseases and the Pediatric Brain Center (Kinderhersencentrum), which strive for implementing innovative therapies at EMC.

Fit with H&T theme(s)

The iMED objectives comfortably fit within the theme Fundamentals of Health & Disease as it connects basic research to health(care) innovations in early detection, disease diagnosis, and precision medicine. iMED aims to **use patient-derived iPSCs to facilitate the development of innovative treatments aimed at correction of genetic deficits with AONs and gene therapy.** Throughout this process, we will **obtain basic knowledge of the genetics underlying rare diseases** and **develop a platform for translation of this knowledge and relevant technology into actionable diagnostic information**. In addition to the development of ethically acceptable, novel therapies, our **platform is ideally suited to use iPSCs to advance fundamental research of human biology in health and disease.** Additionally, the iMED platform serves as a **highly advanced diagnostic tool for undiagnosed rare diseases** when next generation sequencing alone is insufficient to provide a conclusive diagnosis.

2.b Synergy between disciplines

Tailored treatment of individual patients with rare diseases is commercially unattractive to industry, whereas classical drug discovery is generally out of reach for academia. Advances in genetic therapy, however, shift this (dis)balance, in particular because the chemical modifications needed to increase AON stability are recently no longer under patent protection. An iPSC-based platform to develop personalized (genetic) therapies is an essential tool to make the screening of optimal AONs feasible.

The first step towards this platform is establishing an automated iPSC reprogramming and differentiating unit to generate disease models to study the effect of (genetic) therapies. The EMC research groups with expertise in stem cell derivation (Gribnau, Ghazvini) and differentiation (Kushner, De Vrij, Van Woerden, Pijnappel) in collaboration with experts from TUD in industrial design for automation (Goossens), will develop a highly automated iPSC reprogramming and differentiation unit to develop neuronal and neuro-muscular models. To enable assessment of therapeutic efficacy, electrophysiological and optical readouts will be established and integrated by TUD experts in microelectronics (Sarro, Mastrangeli), imaging physics (Brinks), and biophysics (Meijer). EMC experts in AON development (Elgersma, Barakat, Pijnappel, Van Woerden) will come together with TUD experts in Artificial Intelligence (Reinders) to develop algorithms to predict suitability of gene candidates and *in silico* optimization of AON sequence selection. The experience of EUR (Rutten, Hakkaart) in health technology assessment of rare genetic diseases (*e.g.,* first economic evaluation of an orphan disease [Pompe] in the Netherlands) will greatly contribute to cost-effectiveness analyses and health insurance reimbursement associated with bringing the therapies to patients, while ethical and societal questions raised about end-user perspectives will be addressed by EMC (Riedijk).

Through this convergence and close collaboration within the Flagship, strong interdisciplinary connections between **life sciences, engineering, and social sciences** will be established and serve as catalysts for further research in gene therapy.

2.c Long term sustainability plan

Within the flagship funding period, the iMED consortium aims to set up an automated system for upscaled iPSC-derivation, differentiation, and advanced readouts, ultimately enabling identification of disease-correcting AONs within 1 year of a clinical genetic referral. The next step, the actual administration of AONs or gene therapy vectors to the patient, is expected to take place within years after the central funding period. This implementation will take place at the *ENCORE* expertise center for neurodevelopmental disorders (headed by Elgersma), in collaboration with the Rare Disease Innovation Center and the Pediatric Brain Center at the ErasmusMC-Sophia's Children hospital. Ultimately, iMED will help lay the groundwork for a national treatment center for children with rare neurodevelopmental and neuromuscular disorders.

The **feasibility** of iMED is demonstrated by the involvement of partners (Elgersma, Pijnappel) in the translation of pre-clinical discoveries into human clinical trials (through the *ENCORE* and Pompe expertise centers). Currently, an AON trial to restore protein function in the severe neurodevelopmental disorder Angelman Syndrome is ongoing at *ENCORE* (Elgersma). AONs and lentiviral gene therapies have been developed (Pijnappel) for the treatment of classic infantile and late-onset Pompe disease for treating muscle and the CNS.

Several factors contribute to the **long-term sustainability**:

- The PIs have **a strong record in obtaining funding**. The two ZonMW-PSIDER grants and the Organ-on-a-Chip Zwaartekracht grant in which many partners are involved, provide a solid financial and intellectual basis for developing iMED. This is evidenced by the ability of the consortium to match over 1.5M€ in cash for the first two years.

- **The proposal fits well with many National and International research initiatives,** as indicated in **section 3d** (*e.g.* the recently announced 60M€ EU funding call for RNA-based therapies and diagnostics for rare genetic diseases), and hence has the strong potential to attract further funding. External funders and collaboration partners will be engaged during the funding period (see **section 2d**) to allow for continuation after central funding ends and to support extension to other diseases. Since **both iPSCs and genetic therapies are emerging technologies**, we expect numerous funding calls in the (near) future. Establishing the iMED Flagship would make us extremely competitive to receive such future funding. Currently, several applications are in the final round of review such as several NWA applications, NWO-Groot and NWO-Medium applications and VENI and VIDI applications.
- **Partner network expansion and education** (**section 2d**) will also play key roles to ensure self-sustainability and growth. Educational programs ensure the next generation of scientists, engineers and sociologists will understand and contribute to advancements in the field. Academia, industry, and government partnering will promote continued platform development and financing, while collaboration with patient organizations, policy makers and health insurers will ensure successful clinical implementation.

We identified a few **risks** that could impede success of iMED:

- **Technical/Scientific:** Although the platform partly relies on reverse-engineering to ensure interfacing with existing technologies (*e.g.* commonly used multi-electrode recording arrays) there is a considerable challenge involved in designing new technologies that fulfill the needs of iMED. Notably, once the technology is established, it must be kept in working order and expansion of the platform is expected. We envision that liaising with industry partners able to license and produce these chips in large volumes will be required.
- **Financial:** Since our ultimate dream is to offer our therapies at cost price to patients, we participate in the non-profit Dutch Center for RNA Therapeutics (DCRT) (in which Elgersma is board member) and are having discussions regarding the financial aspects with Zorg Instituut Nederland (ZiN) and the Dutch ministry of health (VWS). We are also open to partnering with pharmaceutical companies for common disorders. We have previously successfully partnered with Hoffmann-La Roche and Ionis for preclinical AON development.
- **Regulatory:** Treatments on a 'named-patient' basis ('Artsenverklaring') might not be accepted by the authorities to treat patients with our innovative therapies. In such a case, we will devise a detailed plan to conduct requisite studies for obtaining authorization. Through the aforementioned DCRT, we are liaising with the CBG (Dutch Board for the Evaluation of Medicines) and the EMA (European Medicines Agency) to align with regulatory agencies. The DCRT recently had its first formal discussion with the Innovation Task Force of the EMA.
- **Societal acceptance:** Gene therapy is still marred by the hesitancy of society to genetic intervention. Open dialogue and information sharing through public engagement will increase acceptance by patients and society (see **section 2d**). This will also be facilitated by the new EMC research masters 'Genomics in Society'.

2.d Visibility, attracting talents and new partners

The program will primarily be carried out by **PhD students, supervised by early career postdocs and (co)promotors from two disciplines.** Additionally, iMED is perfectly suited for bachelor and (research) master students (see **section 4a for a comprehensive overview of bachelor and master programs in which the iMEd team currently participates**). Participation of students in iMED will be facilitated by an **Education Team** lead by Van Woerden, as described in **section 4d**.

Early career scientist Van Esbroeck will assist Elgersma with the (scientific) coordination of iMED. Early career PI's Tjakko van Ham (VENI; functional genomics) and Virginie Verhoeven (VENI; and retinal disorders), will join the iMED Flagship in the coming years, as **the iMED platform expands into a tool for functional genetic studies and a platform for identifying genetic treatment for retinal therapies** (lead by Caroline Klaver) (**Figure 1**). For **clinical studies** we will collaborate with Marie-Lise van Veelen (neurosurgeon, head of **Pediatric Brain Center)**, Marie Claire de Wit (neurologist, clinical director of the ***ENCORE* expertise center for neurodevelopmental disorders**) and Ans van der Ploeg (pediatrician, head of the Center for **Lysosomal and Metabolic disorders**).

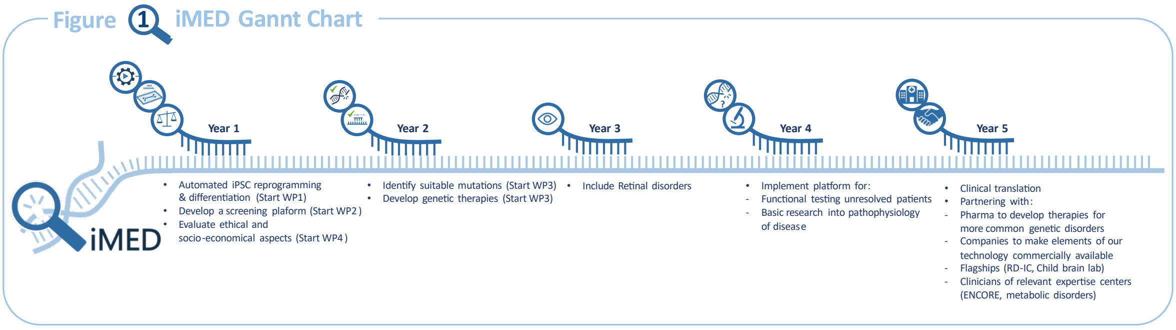


Figure 1. *iMED Gannt Chart, showing start of work packages as well as envisioned start of new projects and iMED alignment with the Flagships Rare Disease Innovation Center (RD-IC) and Child Brain lab.*

Various communication measures will be utilized to increase visibility, attract talent and new partners (table below), and to involve stakeholders such as patient organizations as well as the general public. These activities will be overseen by **the communication and dissemination team** (as described in **section 4d**), led by Riedijk, who was also involved in the Dutch 'DNA dialogue' and director of the **Genomics in Society** masters.

Table 1. *Measures to increase visibility, and attract talent and new partners.*

Target group and objective(s)	Tools and/or channels
Students Stimulate early interest in the field. Generate and strengthen an inflow of talent through internships and PhD programs.	Focused courses within bachelor and masters programs: Medicine, Life Sciences & Technology, Nanobiology, Applied physics, Microelectronic engineering, Mechanical engineering, Biomedical engineering, Clinical Technology, Computer Science, Nanobiology, Molecular Medicine, Genomics in Society, Health Sciences, Neuroscience, Health Economics, Policy and Law, Management.
Research groups Attract additional collaboration from experts in the field.	Top level publications in open access journals Symposia/conferences
End-users (Clinicians and Patient advocacy groups) Obtain feedback to guide our research (as part of ethics-guided research) and to increase engagement.	Stakeholder meetings (through the *ENCORE* and Lysosomal and Metabolic disorders expertise centers), focus groups, social media, website, videos.

Scientific organizations Broaden our network of influence and attract talent and partners for further collaborations.	Presentations at scientific meetings, national and international forums
Industry Explore opportunities for economic valorization of project results and to attract funding.	Social media (*e.g.* LinkedIn), business events, symposia/conferences. Note that many partners are already collaborating with industry partners, including leading pharmaceutical companies in AON and gene therapy technology.
General public Create visibility for iMED and strengthen public acceptance for novel gene therapies.	Social media, website, videos, popular science articles in newspapers, television (talk) shows, TEDx talks, science festivals, and science cafes, and 'Genomics in Society' involvement.
Government and regulatory bodies/Policy makers Facilitate translation of project results through approvals and reimbursements and attract future funding through grant and subsidy applications.	Symposia, direct meetings, science festivals.

2.e Long term societal impact

Healthcare impact:
Brain disorders represent 35% of the total disease burden in Europe with an estimated cost of 800 billion euros in 2010[2]. With the revolution of genomic sequencing over the past two decades, an increasing proportion of brain disorders have been identified as distinct rare diseases with monogenic etiologies. **Monogenic neurological and neuromuscular disorders constitute more than half of all rare diseases.** Collectively, **rare diseases affect >20 million people in Europe** alone[3,4]. **Despite this huge number, they are a disparate group of diseases, with wide biological and clinical diversity, which makes it commercially unattractive to develop therapies, despite the transformative clinical opportunities.**

Through the iMED flagship, **personalized gene therapy can be readily developed in an academic setting** and offered to the patient at cost price. This consortium will provide a platform for technical implementation of genetic therapies in rare diseases, which is initially focused on limited number of diseases to obtain proof of principle, but will be extended to other neurodevelopmental, neuromuscular, and retinal diseases within the 10-year Flagship horizon. We envisage to **treat the first patient** on a 'named-patient'-basis ('artsenverklaring') within **5 years after the launch of iMED.**

Societal impact:
Rare diseases, and certainly brain and (neuro) muscular disorders, can have a strongly adverse effect on quality of life. Such effects are not limited to the patient alone, as typically also families and care givers are affected and ultimately society as a whole. We expect that the iMED platform will greatly enhance quality of life, by enabling the development of personalized treatments. The MSc program 'Genomics in Society' at the EMC, trains researchers to work in an interdisciplinary context, incorporating societal and ethical aspects of genomics for positive societal impact.
Scientific impact:

With the successful implementation of iMED, we expect to dramatically enhance our knowledge of disease pathophysiology. In doing so, we will likely uncover novel insights into fundamental biology and the etiology of rare genetic diseases.

Besides the creation of a platform that encompasses the entire workflow, from iPSC generation and differentiation to therapy development, the various outputs – automated iPSC reprogramming and differentiation, organ-on-chip models, algorithms for identification of suitable mutations and optimization of AON design – will be available for creating further scientific impact through interactions with other basic scientific and clinical research.

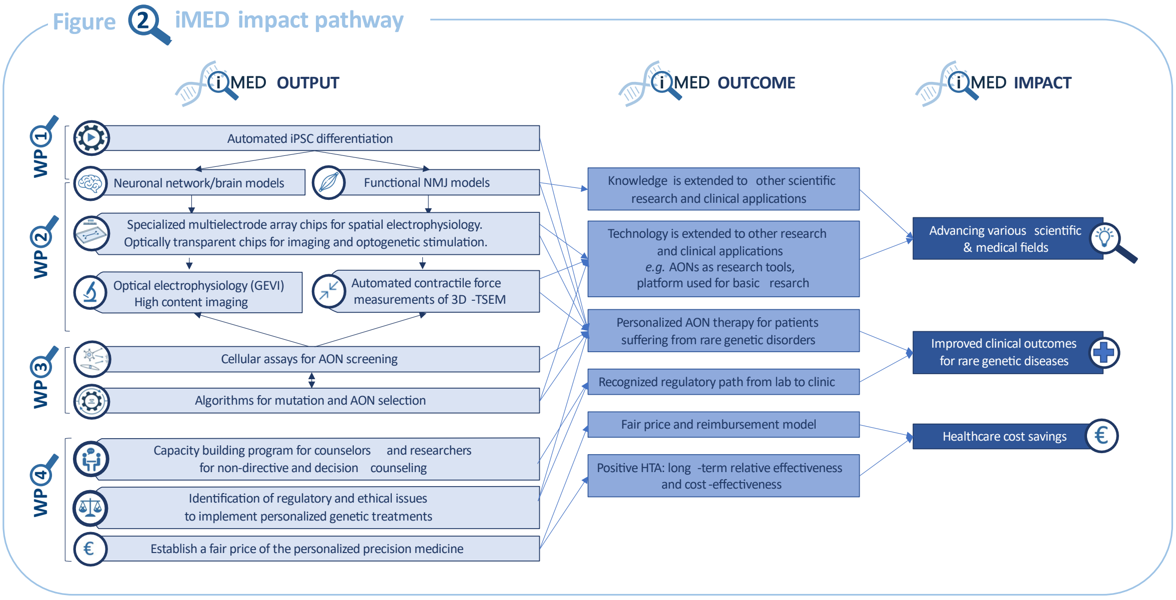


Figure 2. Schematic overview of expected iMED impact.

3. Flagship proposal (25%) (1926/2000 words for 3.a to 3.d)

3.a Ambitions

The **ambition of iMED is to develop a technology platform** that allows us to **identify treatments** for genetic disorders and define a **conceptual, ethical, and economic framework** that enables us to **implement such precision medicines in the clinic**.

The innovative and challenging aspects of iMED are represented at multiple levels. The iPSC platform requires a very close collaboration between TUD and EMC partners to develop methods and technology to automate cell differentiation and to develop functional read-outs for relevant cell types. The biology/genetic challenges require biologists and AI experts to join forces in order to rapidly identify mutations that are amenable to treatment and accelerate the optimization process of AONs or gene therapy vectors. For implementation, we are confronted with very novel ethical and economic questions to prescribe highly personalized experimental drugs to patients. Our partners at the EUR are uniquely qualified to address these complex issues.

We are convinced **this platform will ultimately create a sustainable environment that enables rapid translation of genetic findings into personalized medicine, and place TUD, EUR and EMC in the unique position to be a major leader in this emerging field.**

3.b Aims and feasibility

The high yield of genetic diagnostics provides unique insight into the fundamental biology of genetic disorders. It allows us to develop precision medicine for disorders with a monogenic etiology, specifically aimed at correcting the deleterious effect of the mutation at the RNA or DNA level.
Ultimately, iMED aims to develop an iPSC-based platform to assess functional properties of relevant cell types to facilitate the discovery of (personalized) treatments for rare diseases, and to develop a framework to bring these drugs to patients. Our specific aims are listed in **section 3c** and **Figure 3**.

The **feasibility** of iMED is demonstrated by the **strong scientific and funding track record** of its individual partners relevant to this Flagship, and the **enormous synergy** that rises from their combined expertise. Moreover, AON and gene therapy treatments have already been approved for several rare diseases and **EMC is the world leading center for an AON trial to treat Angelman Syndrome and in the process of developing lentiviral gene therapy for Hunter syndrome.**

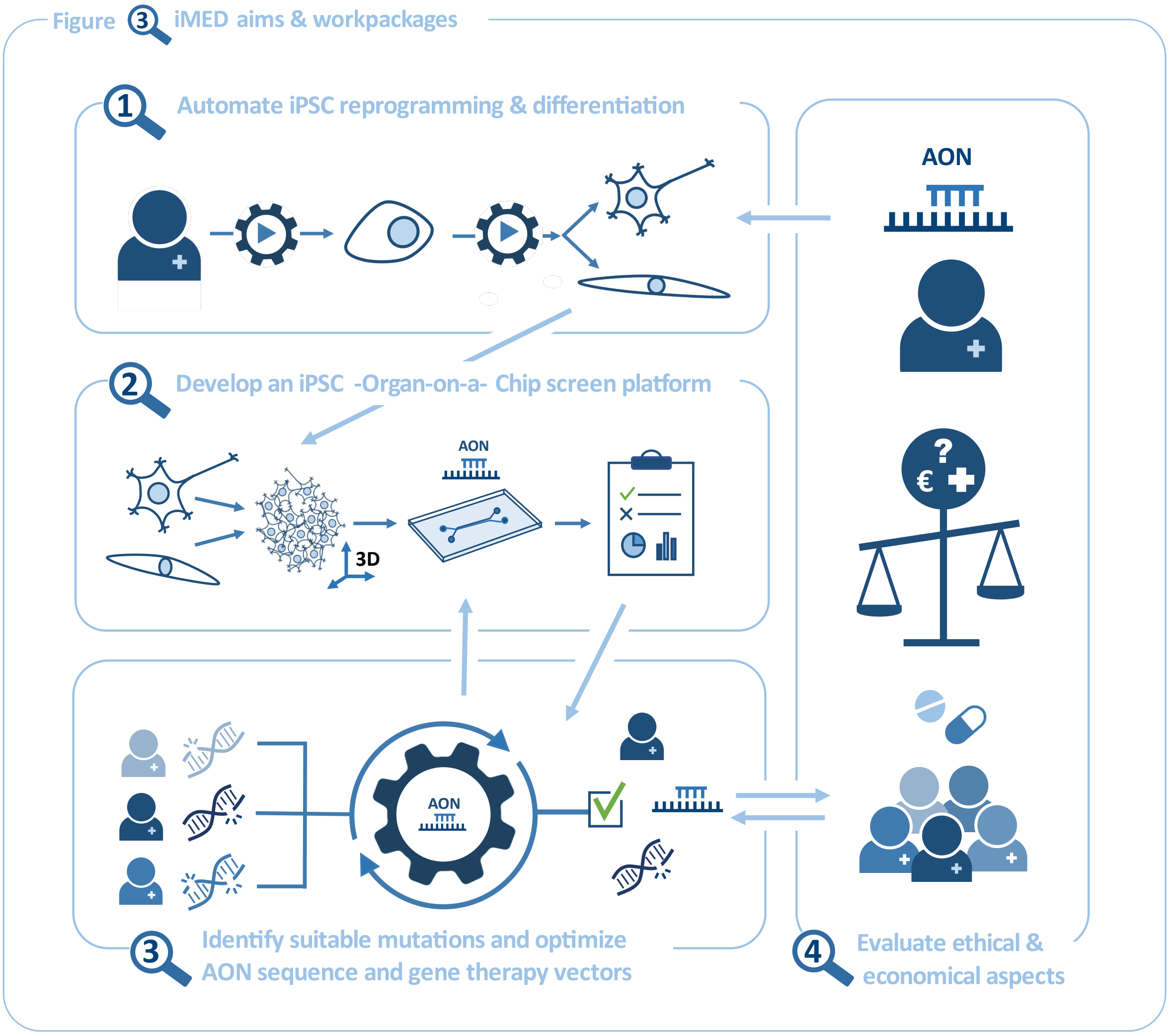


Figure 3. Graphic overview of iMED aims and workpackages.

3.c Work plan

WP1. Establish a platform for highly automated iPSC reprogramming and differentiation

This work package aligns with a pending business case of the EMC iPSC-core facility to build an iPSC-programming street (including a reprogramming unit, expansion unit and freezing unit). Since generation and differentiation of a high number of iPSC clones into relevant cell types and organoids is extremely laborious and susceptible to variability, full automation of the differentiation process is a leap forward to mature the iPSC technology into a platform for functional testing and drug screening[5].

Figure 4 Automation of iPSC reprogramming and differentiation

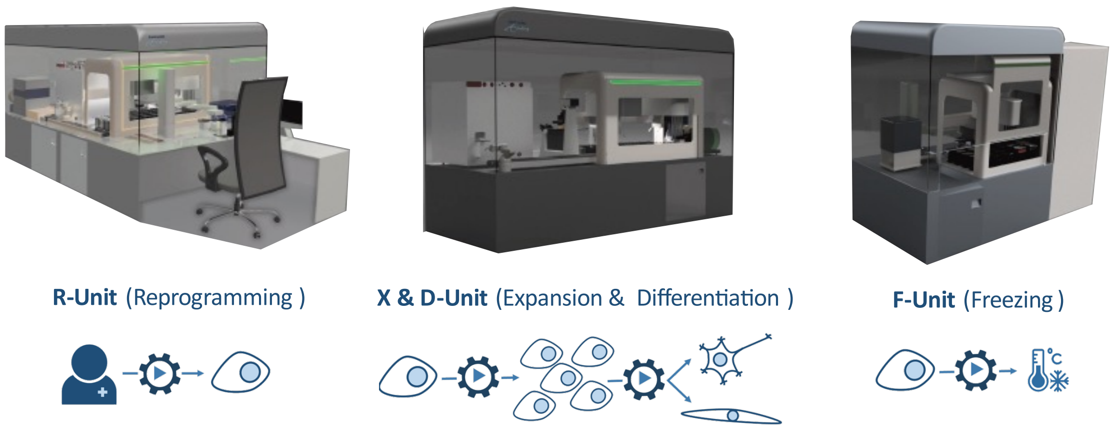


Figure 4. Feasibility study[6] of TUD and EMC to automate iPSC generation, expansion and differentiation, and freezing,

At the end of the Flagship program, we will have:

- Developed protocols for a highly automated differentiation process that is compatible with the X&D unit, to increase sample throughput and to further increase reproducibility (**Figure 4**).
- Developed brain cell culture models ranging from monocultures to fully functional synaptic networks and complex organoid structures recapitulating early neurodevelopment of the cerebral cortex. (**Figure 5**).
- Developed 3D-tissue engineered skeletal muscle (3D-TESM) from myogenic progenitors. Combined with motor neurons generated from the same iPSCs (**Figure 8**), this will yield a functional neuromuscular junction (NMJ) model. Using multi-material microfabrication techniques, we will develop miniaturized platforms for growing these muscle tissues.

Figure 5 iPSC-derived 3D cortical organoids

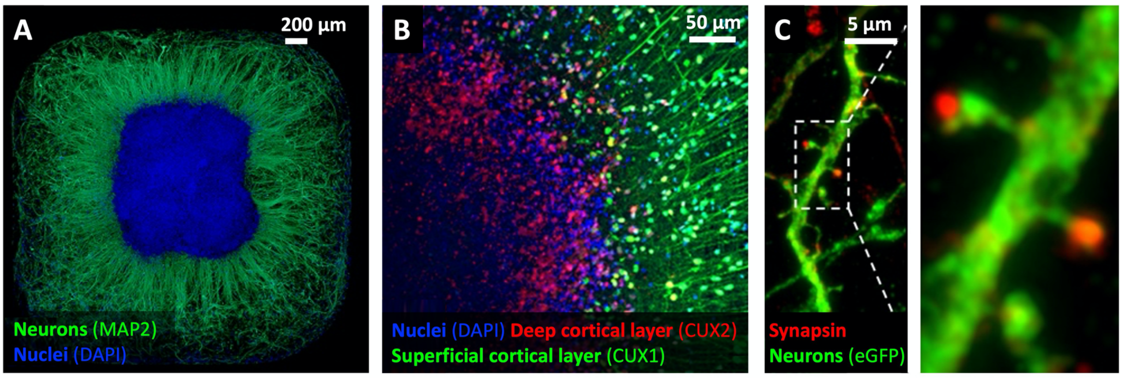


Figure 5. Cortical organoids recapitulate aspects of the developing cerebral cortex. Organoids show radial neuronal outgrowth around a dense core (A), with cortical layering (B), and mature dendritic spines contacting pre-synaptic puncta (C).

WP2. Develop an iPSC-based (organ-on-a-chip) platform with electrophysiological and optical readouts to assess cellular (dys)function.

To enable assessment of therapeutic efficacy, robust functional readouts for cellular function must be established. In WP2 will leverage electrophysiological and optical readouts, ideally in an organ-on-chip-based format, to support increased throughput. Different iPSC-derived neuronal models at EMC[7–11], combined with the development of advanced novel readouts by TUD (nano)engineers, will provide a powerful platform for multi-faceted assessment for phenotype discovery and robust quantification of therapeutic effects of AON treatments developed in **Aim 3**.

In the 2D neural network cultures and the novel adherent cortical organoid platform established at EMC (**Figure 5**), neural network activity will be assessed using a unique combination of standard passive Multi-Electrode Arrays (MEAs) and all-optical electrophysiology with Optogenetic Actuators and Genetically Encoded Voltage Indicators (GEVIs). The integration of customized microelectrodes in planar (2D) and pyramid-shaped (3D) configuration (**Figure 6**) allows for high-resolution spatial interrogation of neural network activity in structurally organized cortical layers. High-content imaging and machine-learning based data analysis of neural networks will be developed with the TUD Bionanoscience department (**Figure 7**).

Figure 6 3D Microelectrode arrays for spatial recording of neuronal activity

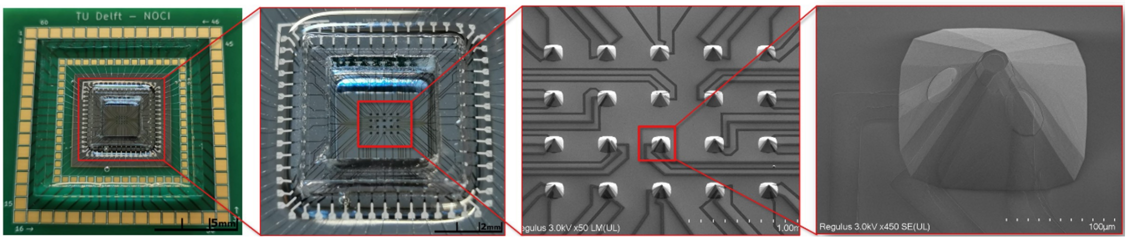


Figure 6. *Array of 3D microelectrodes with multiple independent sampling points distributed in space for extracellular recording of spatial neuronal activity in thick tissues and organoids.*

Figure 7 Adaptive thresholding for fluorescence imaging analysis of neuronal cultures

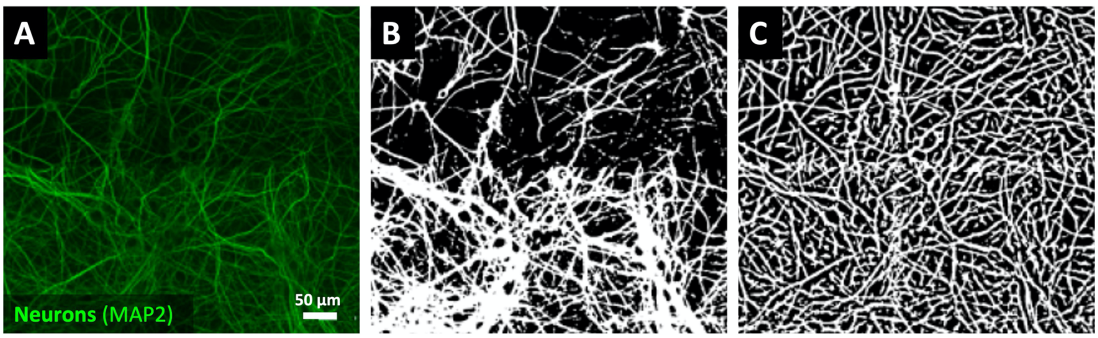


Figure 7. *Fluorescence images of MAP2 neurons show high intensity variability (A) resulting in incorrect thresholding by simple Li-threshold (B) which can be overcome by adaptive thresholding (C).*

Using optogenetics, either the muscle or the motor neuron component of the tissue engineered skeletal muscles (3D-TESMs) can be stimulated with light. This provides a platform to model human neuromuscular disorders *in vitro* and to assess the specific contributions of the different cell types involved. The lithography-based fabrication developed by TUD[12] offers a number of advantages, including the creation of microstructures important for downscaling of platform volume, control over positioning and innervation of motor neurons towards muscle, incorporation of compliant micro-pillars, with the possibility of indentation for consistent positioning of muscle tissue, and integration of pacing electrodes to stimulate muscle tissue. MEA electrodes can be used to monitor neuronal activity and/or muscle activity (EMG) in the same chip. A modular design allows connection of multiple muscle chambers or neuron chambers. Optical transparency enables live force measurement, imaging and optogenetic stimulation (**Figure 8**).

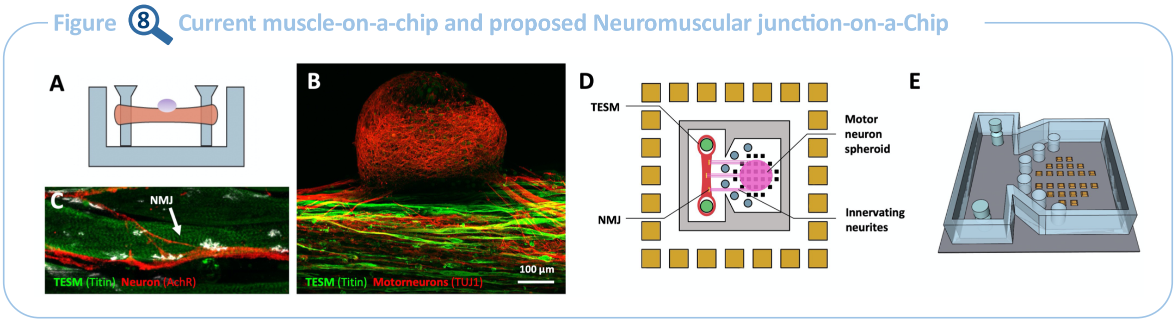


Figure 8. *Schematic representation of the current muscle-on-a-chip (A) and representative immunofluorescent images of 3D-TESM with a spheroid from which motor neurons innervate muscle fiber (B) and the NMJ (C). Proposed NMJ-on-a-chip in 2D (D) and 3D (E), where engineered muscle tissue is grown between two flexible pillars. In an adjacent chamber, the motor neuron spheroid is grown, from which neurites outgrow and are guided to innervate the muscle tissue. A 3D MEA (as in Figure 5) can be devised to monitor electrical activity of the muscle tissue as well.*

At the end of the Flagship program, we will have:

- Developed custom microelectrode chips for neuronal electrophysiological recordings in neuronal cultures and cortical organoids.
- Developed optically transparent chips to allow imaging of organoids and 2D networks.
- Integrated voltage sensors for optical high-content imaging of neural networks.
- The ability to image complete networks at high resolution, to automatically distinguish neurons from glia, and to extract marker parameters, such as axon initial segment length, of all neurons in a high-throughput automated fashion.
- Established a platform for contractile force measurements to measure contraction in tissue engineered skeletal muscles (TESMs).

WP3. Develop *in vitro* and *in silico* tools to assess the amenability of genetic variants to genetic correction and to optimize AON efficacy and selectivity.

As part of the PSIDER program, the Clinical Genetics, Psychiatry and Neuroscience departments have established a list of candidate genes that are of strong interest for genetic (AON) therapy. However, not all mutations are suitable for targeting with AONs, as the efficacy of AONs is highly sequence dependent. Moreover, the identification of the optimal AON sequence and chemistry now largely relies on trial-and-error wet-lab experiments. To facilitate screening, the Clinical Genetics department has developed a cellular bioluminescence-based assay to identify lead AONs, enabling identification of AONs with high sequence specificity for the mutant allele (**Figure 9**). In WP3, this reporter system will be integrated into iPSCs, to fully recapitulate endogenous expression and allow for medium-throughput testing. To further accelerate this laborious AON optimization process, TUD and EMC partners will collaborate to develop machine learning-based algorithms to identify mutations that are amenable AON correction and predict the most effective AON sequence and nucleotide modification *in silico*. We envision that, as we develop and test multiple AONs on our platform, we will be able to further improve these algorithms over time.

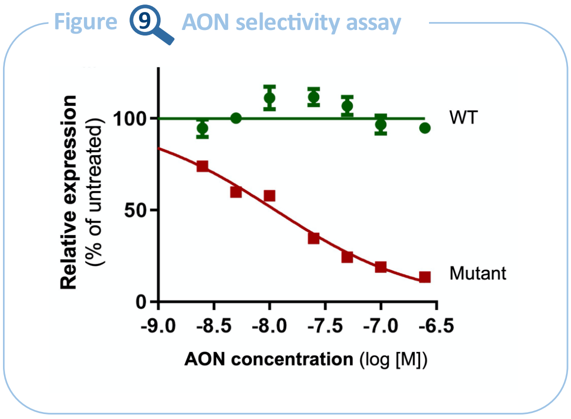


Figure 9. *A bioluminescence-based reporter assay enables assessment of AON efficacy and its selectivity for the mutant allele (here: a single nucleotide mismatch, resulting in selective breakdown of mutant RNA).*

Over the course of this Flagship program, we will have:

- Established cellular assays to screen for effective AONs and gene therapy vectors.
- Developed algorithms and a user-friendly interface that predict goodness of fit of mutations for AON treatment.
- Developed algorithms and a user-friendly interface that predict the most suitable AON

WP4. Establish a framework for an early comprehensive HTA for individually tailored precision medicines to support ethical and socio-economic decision making.

WP4 is dedicated to assessing the socio-economic consequences of early diagnosis and treatment with individually tailored precision medicines by applying an early health technology assessment (HTA). In the early-HTA, we will estimate the life-course medical consumption, societal costs, and health outcomes of patients with unique or very rare mutations for which an AON in iMED can be developed. The early-HTA also addresses ethical, organizational, social, and legal aspects. WP4 is further dedicated to developing a program for counselors and scientist to be enable appropriate counseling and decision-making support to parents.

At the end of this Flagship program, we will have:

- Developed a common approach for the early-HTA of accelerated development of individually tailored precision medicines. Given the rarity of each indication, we will explore whether the unit of evaluation can be shifted from a specific AON or gene therapy for a particular indication towards a treatment based on a particular technology used in multiple indications. We will also explore innovative indicators of value such as ‘value of a cure’, ‘value of knowing’ or ‘value of hope’.
- Estimated the lifetime healthcare and societal costs of patients with rare neurodevelopmental and neuromuscular disorders and compare this to the lifetime costs in case of early-diagnosis and treatment.
- Developed a common approach to establish a fair price of these precision medicines and proposed a suitable payment/reimbursement model in collaboration with payers.
- Established the ethical issues that arise when providing non-registered, individually tailored medicines like AONs and gene therapies to patients. Clear guidelines need to be developed with respect to the information that needs to be provided to fully inform the patient and what information needs to be obtained after treatment is initiated.
- Developed a capacity building program that builds on capacities of clinicians for non-directive counseling of parents; as well as decision-counseling in which clinicians works closely with researchers to support parents in making informed-decisions.

3.d Alignment of the program with relevant national and international research agendas

Our research builds upon and aligns with existing national projects, such as the Netherlands Organ-on-Chip Initiative (NOCI), the NWA-ORC ‘LymphChip’ project, and the ‘Smart OoC’ ‘Perspectief’ project, as well as the proposed Biomedical Production Technology domain of the NXTGEN HIGHTECH program within the Nationale Groeifonds and the European ECSEL-“U "Moore4Medi”al" program. For the iPSC-based work, the flagship aligns very well with the ZonMw PSIDER program, which already funded several iMED members. Since iMEDs use of iPSCs leads to reduced animal experiments, this raises another opportunity for funding (Create2solve program). With respect to translation, iMED perfectly aligns with the ZonMw Translational research program. iMED also aligns with the yearly EU Rare diseases programs, such as the EJP-RD program, which just launched a call for RNA-based therapies this month. With respect to the NWA, we align with multiple topics (*e.g.* route: Personalized medicine).

4. Consortium (25%) (1996/2000 words for 4.a to 4.d)

4.a Track record of the consortium members

The scientific quality of the consortium is reflected by the faculty positions, personal grants, and awards obtained by the applicants (*e.g.* 4xVENI, 3xVIDI, 2xVICI; 2xERC, 2xEMBO-LTF and 3xMarie-Curie Individual Fellowships). In addition, applicants coordinate relevant consortium grants, including ZonMW PSIDER, Zwaartekracht and Horizon 2020. The ability of the participants to match >1.5M€, which is well beyond the required matching for years 1 and 2 (800K€), further demonstrates both the excellent quality and alignment of the consortium members with the iMED Flagship goals.

Most partners are strongly involved in teaching at the Bachelor, Master and PhD level.
BSc: Medicine, Life Sciences & Technology, Nanobiology, Applied physics, Microelectronic engineering, Mechanical engineering, Biomedical engineering, Health Sciences, Computer Science, Clinical Technology.
MSc/MBA: Medicine, Molecular Medicine, Genomics in Society, Neuroscience, Health Economics, Policy and Law, Management, Applied physics, Microelectronic engineering, Mechanical engineering, Biomedical engineering, Computer Science.
Several members are actively involved in **PhD graduate schools**.

At present, the applicants directly supervise more than 25 Postdocs; 55 PhD students; 30 MSc students; 25 BSc students.

Ype Elgersma[11,13–16] (Lead EMC: full professor; VICI) is head of research of Clinical Genetics and founder and director of *ENCORE*. This expertise center has performed multiple translational studies, including an AON trial. Ype coordinates a ZonMW PSIDER consortium.

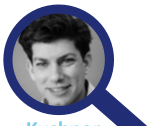
Stefan Barakat[17–21] (assistant professor; EMBO-LTF, VENI, Erasmus MC Fellowship, NARSAD Young Investigator award, KNAW early career award) leads a research group focusing on unravelling the genetics of developmental brain disorders.

Annelot van Esbroeck[8,22–25] is an early career scientist at Clinical Genetics. She has a strong background in life sciences and drug discovery and has affinity with iPSCs to model rare disease

Mehrnaz Ghazvini[7,11,17,26,27] (assistant professor) heads the Erasmus MC iPSC core facility. She is actively involved within ISCI/ISCBI workshop for Good Cell Culture Practice for standardization and quality control of Pluripotent stem cells. She is involved in the national facilities network organizing the annual iPSC user meeting, group leader of CorEuStem Network and International COREdinates Stem Cell Centres.


Gribnau

Joost Gribnau[28–32] (full professor; VICI, ERC) is head of the department of Developmental Biology that founded the Erasmus iPSC core facility in 2010.


Kushner

Steven Kushner[8,11,15,33,34] (full professor, VIDI) is head of research in Psychiatry, and President of the Dutch Society of Neuroscience (Neurofederatie). Steven is founding Coordinator of Brain Research Theme within the Netherlands Institute for Human Organ and Disease Model Technologies (hDMT), PI in the Netherlands Organ-on-Chip Initiative (NOCI) and main applicant of a ZonMW PSIDER consortium.


Pijnappel

Pim Pijnappel[35–39] (associate professor) is board member of the Center for Lysosomal and Metabolic Diseases. His group has developed AONs and lentiviral gene therapy for Pompe disease and MPS II targeting muscle and brain.


Riedijk

Sam Riedijk[40–44] (associate professor) developed and coordinates the MSc 'Genomics in Society' she is editorial board member of the Journal of Community Genetics, and Board member of the Dutch Association for Community Genetics and Public Health Genomics (NACGG).


De Vrij

Femke de Vrij[7,9,11,45,46] (associate professor, ZonMW Create2Solve) successfully established cell culture models for brain disorders. She initiated and leads the EMC iPSC-neuro workgroup and is involved in consortia such as hDMT and NOCI. She teaches in Neuroscience and Nanobiology Master programs and in PhD courses on Regenerative Medicine and Stem Cell Modeling.


Van Woerden

Geeske van Woerden[47–51] (associate professor; VIDI) focuses on calcium signalling-related neurodevelopmental disorders and pathogenicity testing of variants of unknown significance in neurodevelopmental disorders. He has a coordinating role in the BSc Medicine and MSc Neuroscience program.

TUDelft


Sarro

Lina Sarro[52–56] (Lead TUD; full professor) heads the Electronic Components, Technology and Materials Laboratory and specializes in integrated silicon sensors and micro/nano-electro-mechanical technology.


Brinks

Daan Brinks[57–61] (assistant professor, NWO-Startup, ERC starting grant) works at the Imaging Physics department, is PI in Molecular Genetics at EMC, and founder of the Biomedical Intervention Optimization (BIOlab) at TUD.


Goossens

Richard H.M. Goosens (full professor) heads Physical Ergonomics at the Faculty of Industrial Design Engineering.


Mastrangeli

Massimo (Max) Mastrangeli[62–66] (assistant professor) works in the Electronic Components, Technology and Materials laboratory of TUD. Max has expertise in development of microelectromechanical organs-on-chip.

Meijer

Dimphna Meijer[67–71] (assistant professor, VENI, EMBO-LTF, Marie-Curie) leads the Quantitative Neurobiology research group. She initiated the Delft Neuroscience and Engineering work group, that connects young group leaders in neuroscience at TUD and EMC.

Reinders

Marcel Reinders[72–76] is director of Deflt Bioengineering In statute and head of pattern recognition and bioinformatics, which harbors expert research groups on machine learning, deep learning, and bioinformatics.

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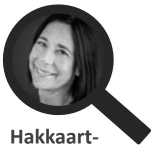

Hakkaart-van Rooijen

Leona Hakkaart-van Rooijen[77–81] (Lead EUR) is associate professor in Economic Evaluations in Health Care, International program director of 'European in Health Economic and Management' (Eu-HEM) and course coordinator master thesis 'Health Economic Policy and Law' (HEPL).

Rutten-van Mölken

Maureen Rutten-van Mölken[78,82–85] is full Professor of Economic Evaluation of Innovations for Health, Head of the HTA department of the Erasmus School of Health Policy and Management, and scientific director of the Institute for Medical Technology Assessment (IMTA).

4.b Expertise of the consortium members in relation to the aims and objectives of the program

WP1

Richard Goossens' research group focuses on ergonomics/human factors in relation to product innovation in order to establish safe, comfortable and error-free task fulfilment in complex multi-user situations in a medical setting. He will be involved in design requirements for the automated iPSC reprogramming and differentiation platform, together with **Joost Gribnau** who is an expert in epigenetic mechanisms with regard to stem cell biology, development, and differentiation. For functional implementation of the automated iPSC platform in iMED, **Mehrnaz Ghazvini (WP1 Lead)** will provide consultation on experimental design, derivation of iPSC lines, and hands on training in differentiation, together with **Femke de Vrij** and **Steven Kushner** who have established diverse human iPSC-derived neural cell culture techniques within their laboratories, and **Pim Pijnappel** for neuromuscular differentiation.

WP2

WP2 requires a very close collaboration between most iMED TUD and EMC applicants. On the technical part, **Lina Sarro (Lead TUD)** is internationally recognized for her work on integrated silicon sensors and micro/nano-electro-mechanical (MEMS/NEMS) technology. Together with **Massimo Mastrangeli (WP2 Lead)** she provides expertise in microfabrication of optically transparent organ-on-chip devices with integrated electrical, mechanical, and chemical. Their main contribution lies in the development and implementation of sensors and functionalities, tailored to the required information acquisition, using a silicon/polymer platform, with potential capabilities of high reproducibility, accuracy, and throughput. **Daan Brinks** specializes in nonlinear imaging (*e.g.*

multiphoton imaging in 3D tissues), voltage imaging and optogenetics. His lab works on the entire pipeline from creation of imaging setups, development of molecular sensors, application in tissue culture models to data analysis and interpretation. **Dimphna Meijer** has expertise in neuronal network formation using molecular and cellular biophysics. She has expertise in high resolution protein structure determination using cryo-EM, X-ray crystallography and SAXS, biochemistry (protein-protein- and protein-DNA interactions) and functional cellular studies with automated high-content confocal microscopy. On the more biological part, **Pim Pijnappel**'s lab developed technology for generating skeletal muscle from iPSCs and generating patient-derived contractile 3D-tissue engineered neuromuscular disease models. He leads several studies on organ-on-a-chip technology and therapy development. **Geeske van Woerden**'s lab focuses on neuronal calcium signalling. Her lab has setup the multi-electrode array (MEA) technique to study formation of neuronal networks. **Stefan Barakat, Steven Kushner**, **Femke de Vrij**, **Ype Elgersma** and **Annelot van Esbroeck** identified and characterized multiple new genetic syndromes, in which they used iPSC- and pluripotent stem cell-based disease modelling for brain disorders.

WP3

Several EMC researchers have a strong track record in translational medicine studies. Recent research by **Ype Elgersma** focuses on iPSCs and AON treatment, in collaboration with **Steven Kushner** and **Femke de Vrij.** Ype is involved in multiple translational trials and currently the first AON trial for the severe neurodevelopmental disorder Angelman Syndrome, is performed at EMC. **Annelot van Esbroeck** worked on drug screening assays and activity-based protein profiling, including the use of iPSCs together with **Steven Kushner**'s lab. **Pim Pijnappel**'s (**WP3 lead**) group has developed AONs for late-onset Pompe disease and lentiviral gene therapy for classic infantile Pompe disease and MPS II targeting muscle and brain. **Marcel Reinders** is experienced in applying (novel) machine learning techniques to develop complex bioinformatics algorithms. Specific expertise related to WP3 deals with predicting pathogenetic effects of mutations using sequence context of the mutation.

WP4

iMED has a very strong team of researchers with expertise in ethical and economical evaluation studies, and the engagement of the public. **Maureen Rutten-Van Mölken**'s **(WP4 lead)** expertise includes economic evaluation studies, either alongside clinical trials or using probabilistic decision-analytic modeling and outcomes research. She has a special interest in the cost-effectiveness and the financing and payment of complex multi-faceted interventions. **Leona Hakkaart-van Roijen (EUR lead)** is experienced in the field of economic evaluations in health care. She was co-promotor of the first economic evaluation of an orphan disease (Pompe) in the Netherlands, and author of the costing manual, part of the guidelines for economic evaluations in Health care commissioned by the National Health Institute (ZiN). She is WP-Lead in EU-'PECUNIA' that established standardized costing and outcome assessment measures for optimized national healthcare provision in the European Union. **Sam Riedijk** conducts research into the impact of technological advances in genetics on individuals, families, and society at large. This includes informed decision making and dealing with uncertainty about genetic test results. In addition, her group is an international frontrunner in developing the expertise to engage a broad diversity of publics in dialogue about human germline gene editing (HGGE). She heads the Msc '*Genomics in Society*', which is ideally suited to involve the public in this technology debate. The students of WP4 will be co-supervised by **Ype Elgersma** who, as head of the *ENCORE* expertise center, has been closely involved in translational studies.

4.c Involvement of partners

At present many of the applicants of iMED have active collaborations with external partners that are relevant to this Flagship (see **Figure 10**). During the Flagship program, we will continue to build upon existing partnerships and create new ones.

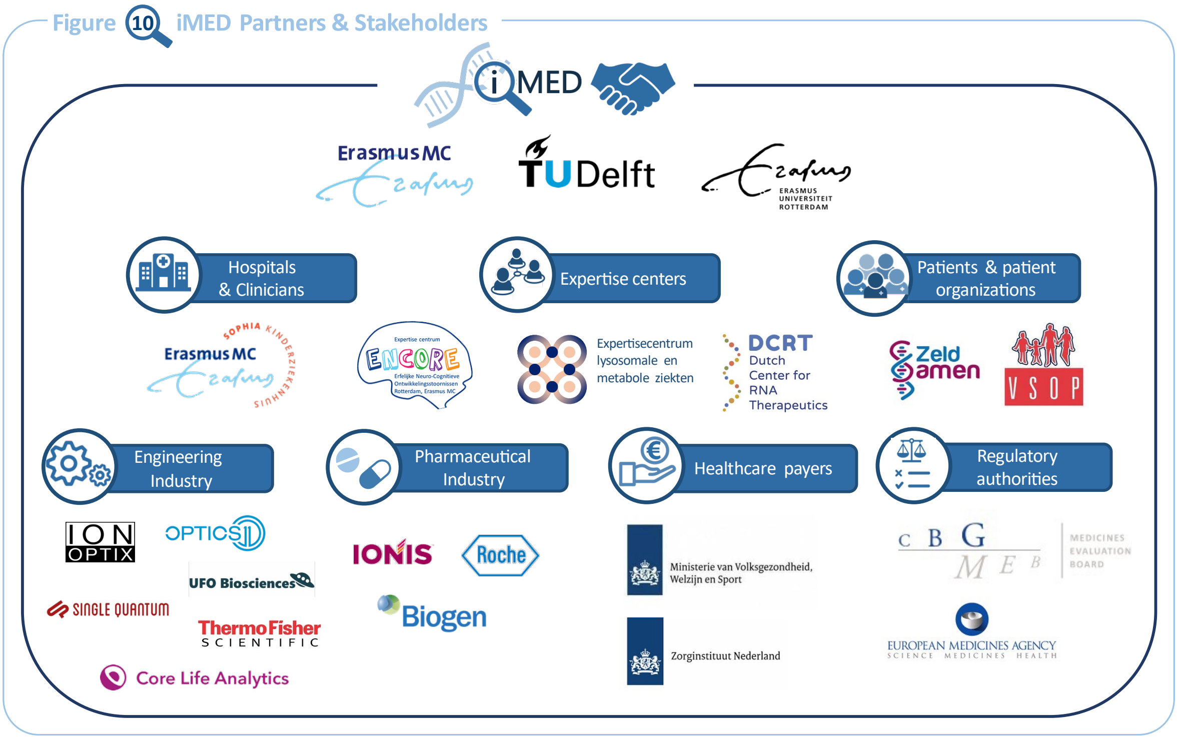

Figure 10. iMED partners and stakeholders

4.d Governance, organization, and collaboration strategy of the consortium

For successful implementation of the iMED project, we will incorporate a clear and transparent organizational structure *(Figure 11)*.

- The **iMED Steering Committee** will consist of representatives from all 3 knowledge institutions (Project Leader, Institute Leads, WP Leads). The project leader chairs the yearly committee meetings. The Steering Committee will direct the overall strategy towards achieving project objectives, monitor overall progress and decide contingency plans. It will also ensure strategic alignment of iMED with the goals of Convergence.
- iMED will be managed by the **project management team**, consisting of the **Project Leader** (Elgersma), and an early career researcher **Scientific Project Manager** (Van Esbroeck). The Scientific Project Manager will support the Project leader by coordinating projects between the institutes and across WPs, organize iMED Consortium Assembly meetings. Administrative responsibilities will be delegated to a secretary of the Project Leader.
- All applicants will jointly form the **iMED Consortium Assembly**. The assembly will be chaired by a rotating WP leader. The Consortium Assembly will meet every 6 months to discuss progress, finances, and interdisciplinary collaboration.
- The **WP Leaders** will supervise their respective **iMED research teams** and coordinate collaborations between various participants.
- iMED will also have a **Communication and Dissemination team**, and an **Education team**. The former will be led by Riedijk who has extensive experience in public outreach and organizing stakeholder and general public discussions. The latter will be led by Van

Woerden, who holds a 0.6 fte teaching position and has a University Teaching Qualification. The Education team will organize symposia, colloquia with external keynotes, facilitate joint supervision for PhD and postdoc projects from at least two institutes, and involve BSc and Msc students from all relevant programs. Both teams will comprise of senior experts who will act as mentors for other members of the team and early career researchers and postdocs who will ideate on additional activities to converge and provide input on how to optimize interdisciplinary working conditions.

- **iMED Advisory board:** External advisory board will be established to provide input and advice to the Steering Committee. Scientific Advisors will consist of internationally renowned experts and key opinion leaders as well as ethicists, legal advisers, and stakeholders such as patient groups/patient representatives and clinicians. The Advisory Board will meet once a year with the iMED Steering Committee.

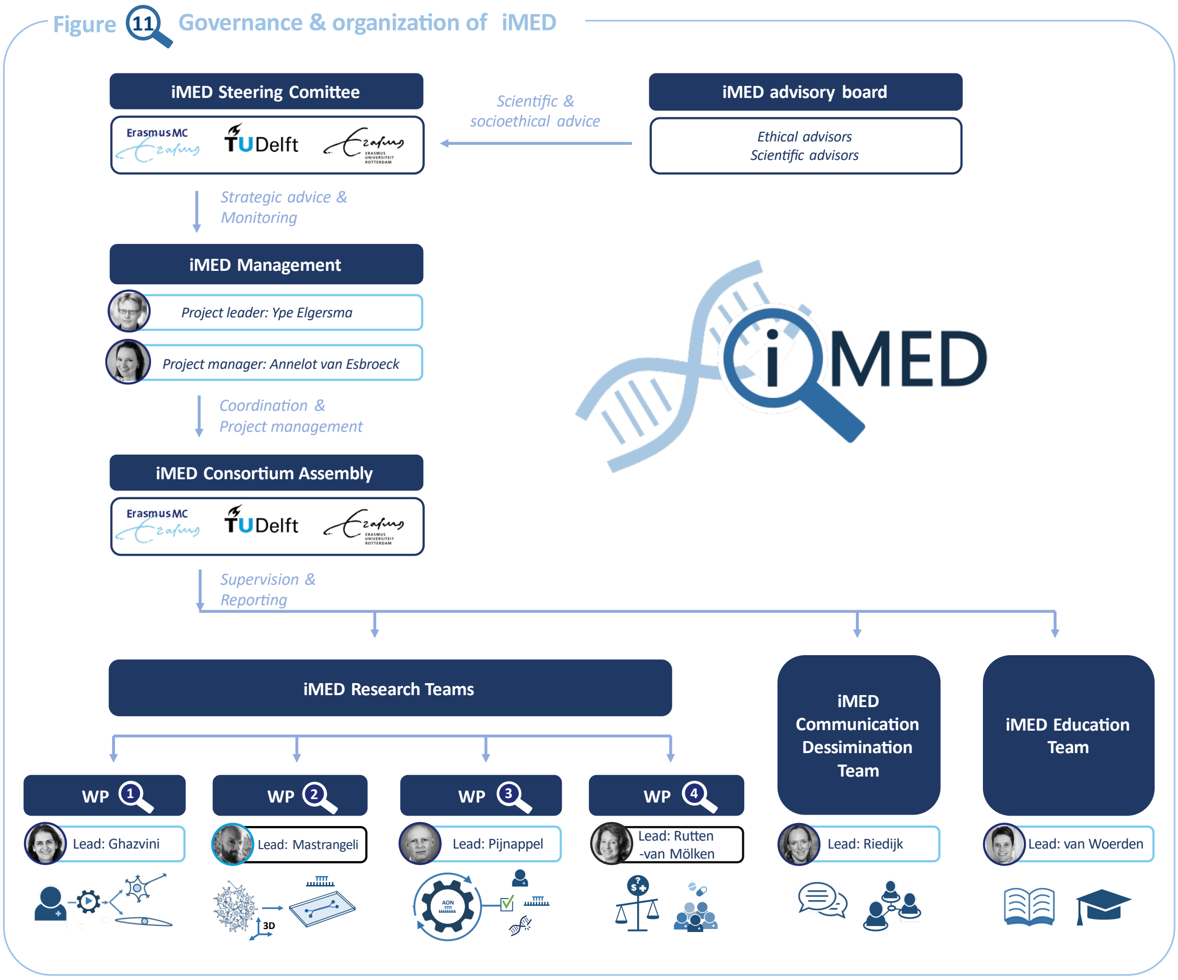

Figure 11. iMED organization.

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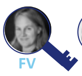

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5. Annex II. Letter of Support

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Head of Research and Education
Prof. Y. Elgersma, PhD

Concerning Letter of support for the Convergence 'iMED' FLAGSHIP

To whom it may concern,

In my capacity of Interim Head of the Department of Clinical Genetics, Erasmus MC, I want to express my full support for the Flagship Convergence project **'An iPS based platform for developing personalized medicine for rare diseases (iMED)',** as applied for by the main applicant, Ype Elgersma, Chair of Research and Education of the Dept. of Clinical Genetics at Erasmus MC.

The research of Clinical Genetics would tremendously profit from the IPSC-based platform developed under *iMED,* as it fully aligns with the strategic research focus of the department to not only provide a genetic diagnosis, but also assess the possibility for therapeutic correction. In particular, the emergence of innovative genetic technologies brings this possibility a large leap forward.

In addition to a drug screening platform, the *iMED* iPSC-based platform will also be a very powerful tool for studying the pathogenicity of genetic variants of unknown significance in the relevant cell type, and hence also contribute to our diagnostic care.

The long-term viability of iMED consortium is warranted by the strong track record of the applicants (who are able to provide matching well beyond the required amount), the good alignment with National research programs such as 'Zwaartekracht-'Organ on a chip' and 'PSIDER' in which many applicants participate, and the Erasmus MC-Sophia focus areas 'Rare diseases' and 'Kinderhersencentrum'.

Yours sincerely,

Prof. P.A.E. Sillevis Smitt, MD, PhD
Interim Head of Department Clinical Genetics

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